

# University Travel & Commuting Analysis

*Booked business travel (flights, rail, car, hotels) plus a staff/student commuting survey, combined into one 12-month travel-emissions picture.*

## Executive summary

This synthetic case demonstrates the workflow from raw operational data to a management-ready analysis. All figures are illustrative and not real client data, and all figures below come from a single combined dataset, a 60-trip illustrative business-booking log and a 60-respondent commuting survey, covering the same 12-month period.

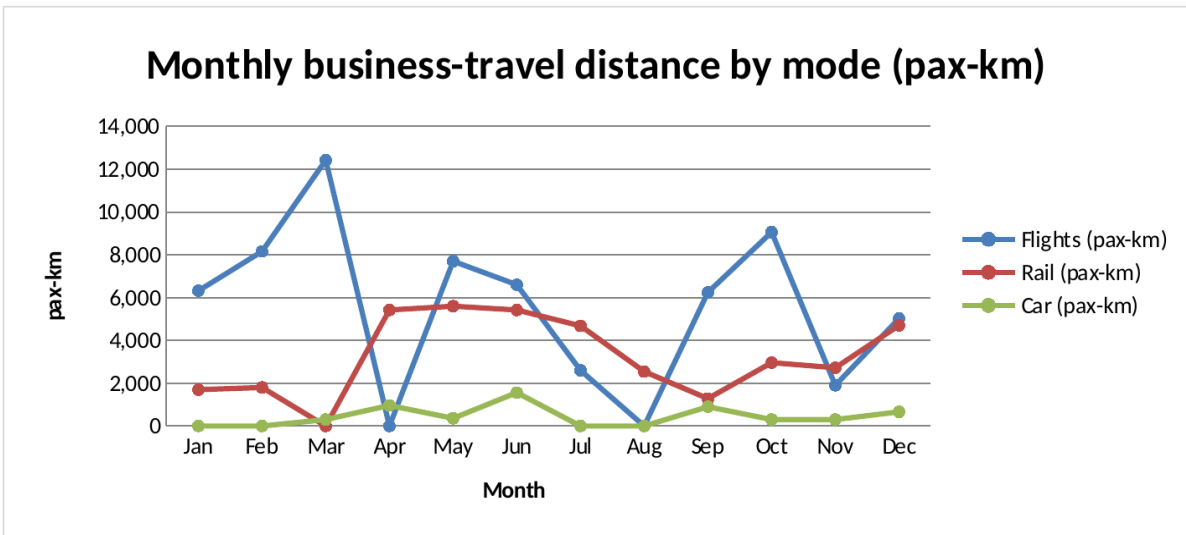
| KPI                                 | Illustrative result | Unit               |
|-------------------------------------|---------------------|--------------------|
| Total travel emissions              | 429.4               | tCO <sub>2</sub> e |
| — of which business travel (booked) | 15.8                | tCO <sub>2</sub> e |
| — of which commuting (surveyed)     | 413.6               | tCO <sub>2</sub> e |
| Commuting share of total            | 96.3                | %                  |

## Method: two data sources, one dataset

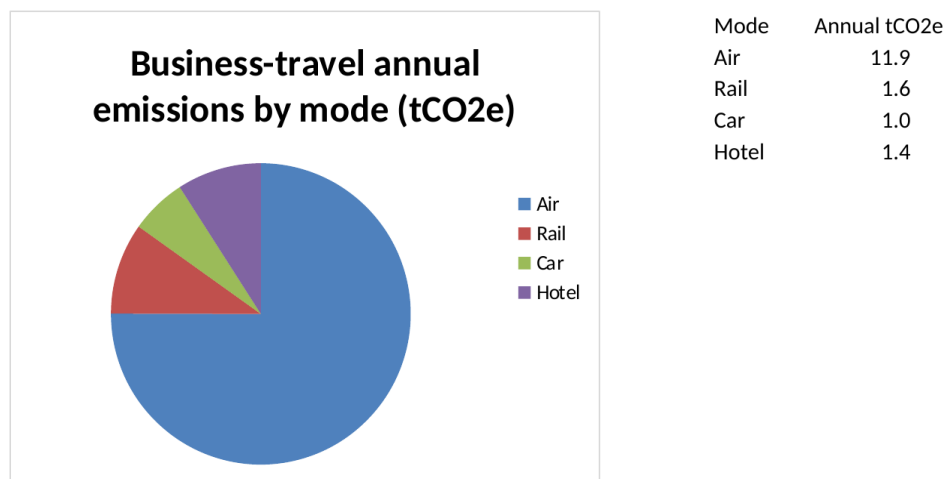
Business travel has a booking trail: a trip-by-trip log naming destination, date and mode, with distance looked up from a destination table. Everyday commuting has no booking trail at all, so it is estimated instead — a short staff/student questionnaire captures a distance band, usual mode and days on campus per week, which are then converted into monthly km assumptions and scaled by each month's occupied FTE. Both streams are aggregated to the same monthly, same-unit (tCO<sub>2</sub>e) basis below, so they can be read side by side rather than as two separate reports.

## Business travel activity

60 illustrative bookings across the year show the expected term-time pattern: trip volume is highest in the spring and autumn teaching terms and drops sharply in July/August. Distance is looked up from a 10-destination reference table, not typed per trip, so it stays consistent as new trips are logged.



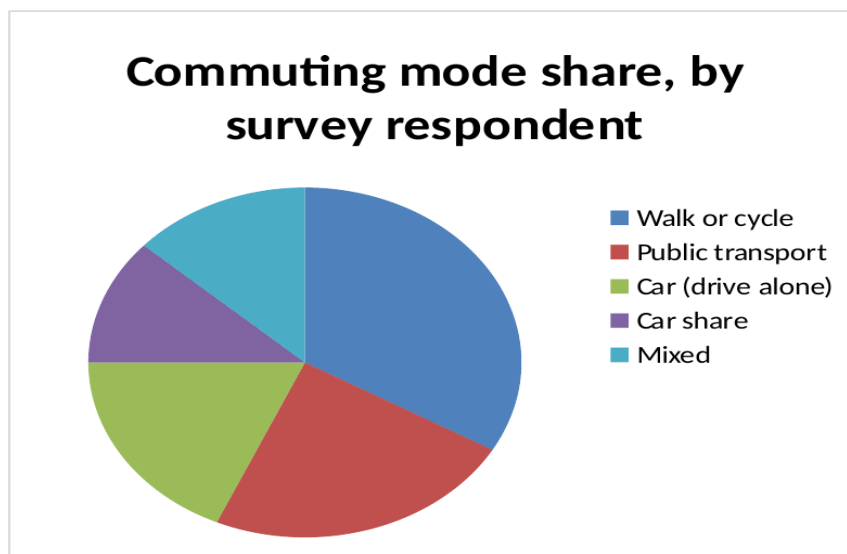
Air travel accounts for roughly three-quarters of business-travel emissions 75.0% in this sample, despite being only one of three modes booked, and despite the much smaller trip volume than a full-scale booking export would show. The pattern holds regardless of scale: a handful of flights outweighs many more rail and car trips.



Key insight: air travel drives business-travel emissions even at small sample sizes — in this illustrative log, 3 long-haul flights contribute more than the other 57 trips combined. A travel policy that targets flight bookings specifically (route consolidation, rail-first for sub-500 km trips, virtual-meeting defaults) reaches most of the reducible emissions here.

## Commuting activity

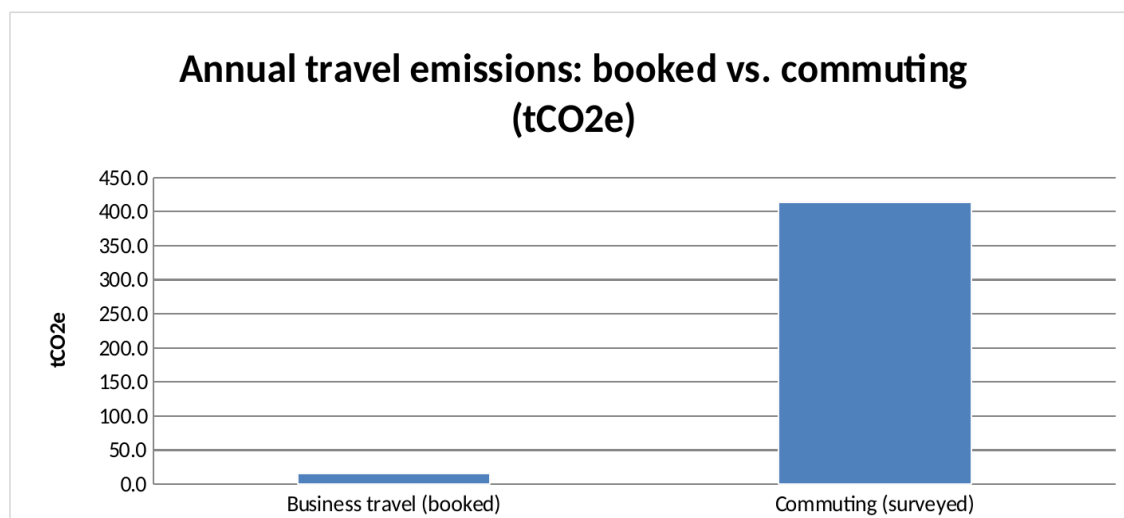
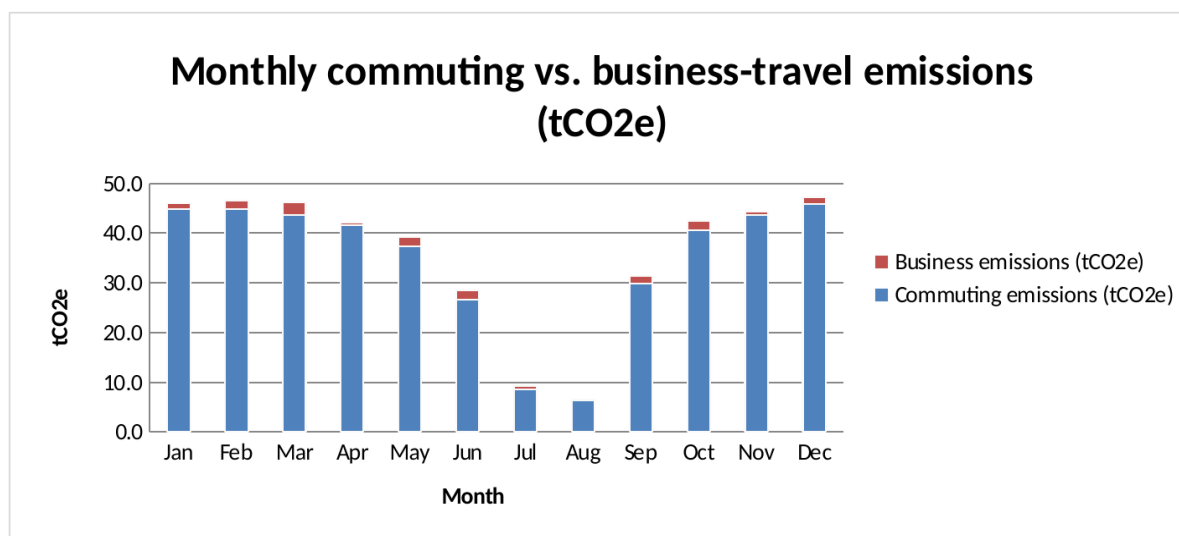
No booking system records how staff and students get to campus, so this is estimated from the 60-response commuting survey (see the companion questionnaire workbook): each respondent gives a distance band and usual mode rather than an exact figure, because that is what people can answer reliably. Walking/cycling is the single largest mode share, followed by public transport and driving alone.



Car-based commuting (drive-alone plus car-share) covers 30% of respondents and carries almost all of the commuting emissions total below, since walking/cycling is treated as zero-emission screening and public transport carries a far lower per-km factor.

## The measurement gap: commuting vs. business travel

Put on the same monthly, same-unit basis, commuting dwarfs business travel every month of the year — including the summer term break, when occupied FTE (and therefore commuting) falls sharply but business travel does not fall nearly as far.



Key insight: commuting accounts for 96.3% of total travel emissions in this combined dataset vs. 3.7% for booked business travel. This matches a well-documented pattern in real university carbon footprints where commuting is high-frequency and invisible to any booking system, while business travel is low-frequency but fully visible. A carbon report scoped to booking data alone would be measuring the smaller of the two categories by a wide margin.

## Data quality

- Commuting figures depend on self-reported distance bands and mode, not metered data — the distance-band lookup (tab 03 of the companion workbook) is kept as a visible, editable assumption rather than buried in a formula.
- Business-trip distances come from a 10-destination lookup table, not typed per trip; a new regular destination needs adding there before the monthly totals will include it correctly.
- Occupied FTE is reused from the building-operations sample for internal consistency across this portfolio; a real engagement would use the client's own enrolment/attendance figures month by month.
- The two data sources update on different cycles (bookings continuously, survey annually/termly) — reconcile reporting periods before combining them for an external disclosure.

## Recommendations

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1. Repeat the commuting survey annually or termly to track mode shift over time. Currently, commuting is 96% of the total, so small changes here matter more than business-travel changes.
2. Target business-travel policy at flights specifically (route consolidation, rail-first below ~500 km, virtual-meeting defaults). That is where most of the reducible business-travel emissions sit.
3. Validate the distance-band assumptions against postcode-level data if/when available, rather than self-reported bands alone.
4. Extend the commuting questionnaire to capture EV/hybrid ownership once car-commuting emissions become a reduction target.
5. Report business travel and commuting as two clearly labelled line items, not a single blended total they come from different data sources with different confidence levels.

Scope note: This portfolio demonstrates data preparation, screening calculations, KPI development and reporting. It is not independent GHG verification or assurance, a statutory energy audit, legal advice, or a guarantee of savings. All figures in this document are illustrative, synthetic data for demonstration purposes only.